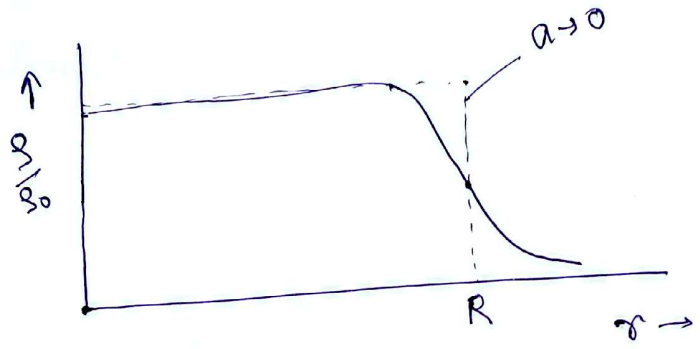
Nuclear density distribution $\rho(r)$

$$\rho(r) = \frac{\rho_0}{\exp\left\{\frac{(r-R)}{a}\right\} + 1} \quad \text{——(1)}$$

$R = R_0 A^{1/3}$ with $R_0 \approx 1.3$ fm $1\,fm = 10^{-15}$ m

A = mass number $= N + Z$ (N: neutron number, Z: proton number)



Cross section for structure less nuclie → σ_{Mott}

Cross section of actual nuclie → $\sigma = \sigma_{Mott}\,[F(q)]^2$ ——(2)

where Form factor $F(q) = \frac{4\pi}{q}\int_0^\infty \rho(r)\,\sin(qr)\,r\,dr$ ——(3)

$q = \frac{2E}{\hbar c}\sin\frac{\theta}{2} = \frac{2}{\lambda\!\!\!^{-}}\sin\frac{\theta}{2}$ ——(4)

(5) —— $\lambda\!\!\!^{-} = \frac{\hbar c}{E}$ is reduced de-Broglie wavelength of electron (probing nuclie)

Normalization condition of charge density

$$\int \rho(r)\,4\pi r^2 dr = Ze \quad \text{——(6)}$$

For $qr \ll 1$ or $\frac{r}{\lambda\!\!\!^{-}} \ll 1$ (means $\lambda\!\!\!^{-} \gg$ nuclear scale r)

$\sin qr \approx qr - \frac{1}{6}q^3r^3 + \dots$

Then Eq. (3) become

$$F(q) = \frac{4\pi}{q}\int \rho(r)\left[qr - \frac{1}{6}q^3r^3 + \dots\right]r\,dr$$

From the relation $e^{i\theta} = \cos\theta + i\sin\theta$ in binomial expansion,

$$F(q) = \int \rho(r) 4\pi r^2 dr - \frac{1}{6} q^2 \int r^2 \rho(r) 4\pi r^2 dr + \cdots$$

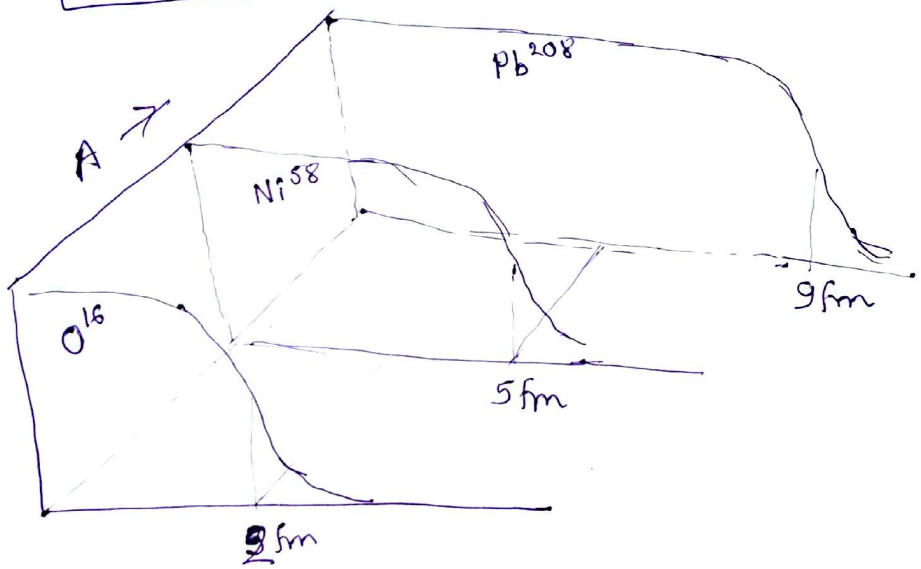
$$= Ze \left[1 - \frac{1}{6} q^2 \langle r^2 \rangle + \cdots \right]$$

Total electric charge Ze can be defined as

$$Ze = \lim_{q \to 0} F(q)$$

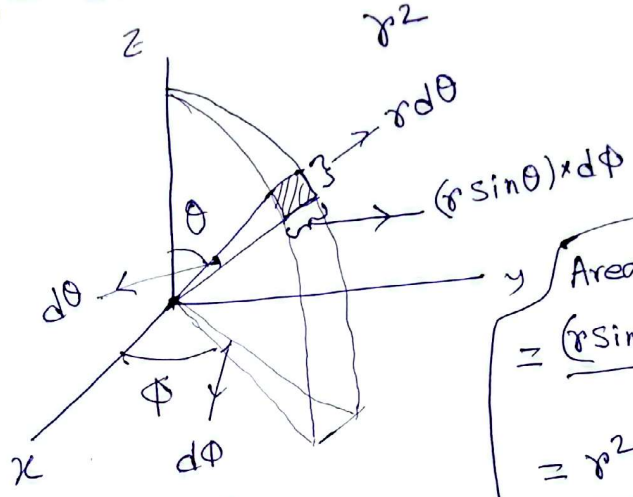
and mean square charge radius can be defined as

$$\langle r^2 \rangle = -6 \left(\frac{\partial F}{\partial q^2} \right)_{q^2 = 0}$$



Quantum description of scattering theory :

Solid angle $d\Omega = \frac{(r\sin\theta)(rd\theta)(d\phi)}{r^2} = \sin\theta\, d\theta\, d\phi$



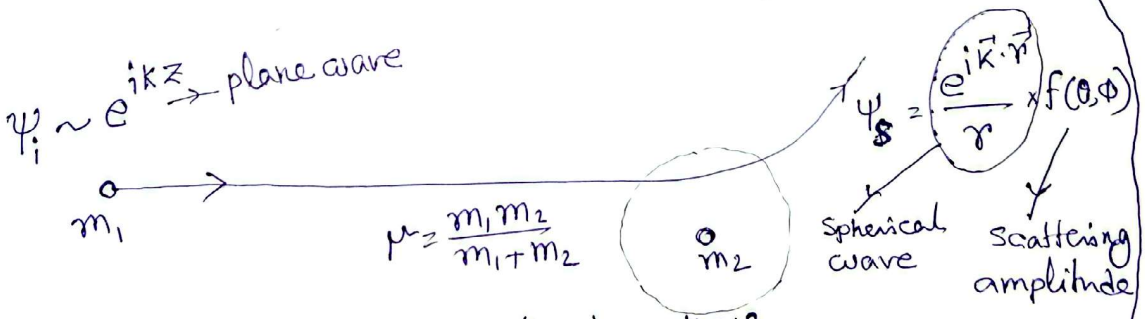
Area/solid angle $= \frac{(r\sin\theta\times d\phi)(rd\theta)}{d\Omega} = r^2$

$$\int d\Omega = \int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\phi = 4\pi$$

Differential scattering cross section $\rightarrow \frac{d\sigma(\theta,\phi)}{d\Omega}$

Total cross section $\sigma_{tot} = \int_{\Omega} \left(\frac{d\sigma}{d\Omega}\right) d\Omega$

$$= \int_0^{\pi} \sin\theta\, d\theta \int_0^{2\pi} d\phi \left\{\frac{d\sigma(\theta,\phi)}{d\Omega}\right\}$$



Initial current density $J_i = \left(\frac{\hbar K}{\mu}\right) \times |e^{ikz}|^2$

(velocity) × (probability density)

or scattered final, " " $J_s = \left(\frac{\hbar K}{\mu}\right) \times \left|\frac{e^{ik\cdot r}}{r} f(\theta,\phi)\right|^2$

$$\frac{d\sigma}{d\Omega} = \frac{J_s \text{ (no. of scattered particles/area/time)}}{J_i \text{ (no. of incident particles/area/time)}} \times \text{(area/solid angle)}$$

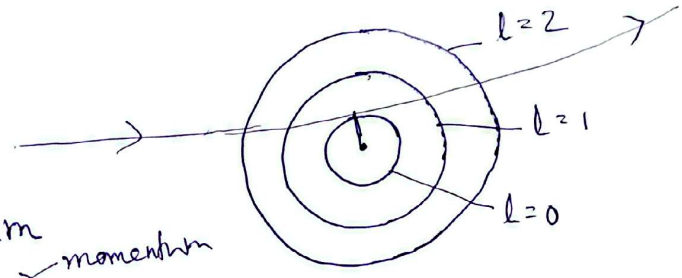
$$= |f(\theta)|^2$$

Solving Schrodinger's equation,

$$f(\theta) = \frac{1}{K} \sum_{l=0}^{\infty} (2l+1)\, e^{i\delta_l} P_l(\cos\theta) \sin\delta_l$$

$$\frac{d\sigma}{d\Omega} = |f(\theta)|^2$$

$$\Rightarrow \boxed{\sigma = \int_{\Omega} \frac{d\sigma}{d\Omega} d\Omega = \frac{4\pi}{k^2} \sum_{l=0}^{\infty} (2l+1) \sin^2\delta_l}$$



Angular momentum

$l\hbar = \vec{r} \times \hbar\vec{k}$ → momentum

↓ potential range

$$\Rightarrow \boxed{l = rK}$$

For nuclear scattering $r \sim$ fm and for few MeV momentum $p = \hbar k$, $rK < 1 \Rightarrow l = 0$ consideration is sufficient, no need to consider $l = 1$ onward contribution

So, $$\boxed{\sigma \simeq \frac{4\pi}{k^2} \sin^2\delta_0}$$

For small momentum K, phase shift $\delta_0 \approx Ka$ can be defined, where a is scatting length.

We will get $\boxed{\sigma = 4\pi a^2}$ as $K \to 0 \Rightarrow \delta_0 \to 0 \Rightarrow \sin\delta_0 \approx \delta_0$

↑ Hard sphere scatting approximation

$\sigma \approx$ mb

$a = ?$

